

01 / WHO'S BUILDING

FULL STACK DEVELOPER (MERN) | UI UX

ANURAG A PANDIT



VISIT MY PORTFOLIO

Hi there, I'm a Full Stack Dev who loves turning ideas into interactive cool web experiences. I excel in MERN with Python, C, C++, DSA and UI/UX with Figma in toolkit



FULL STACK DEV

UI/UX DESIGNER

TECH ENTHUSIAST

// an honest guide for first-time builders

WEB DEVELOPMENT

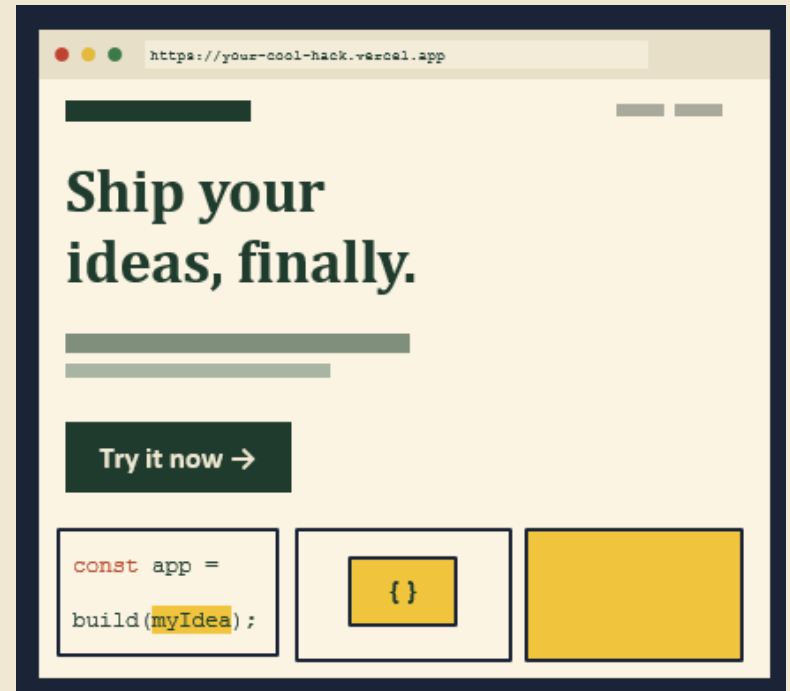
for hackathons – *Anurag Pandit*

From `index.html` to a working product.

"I have an idea."



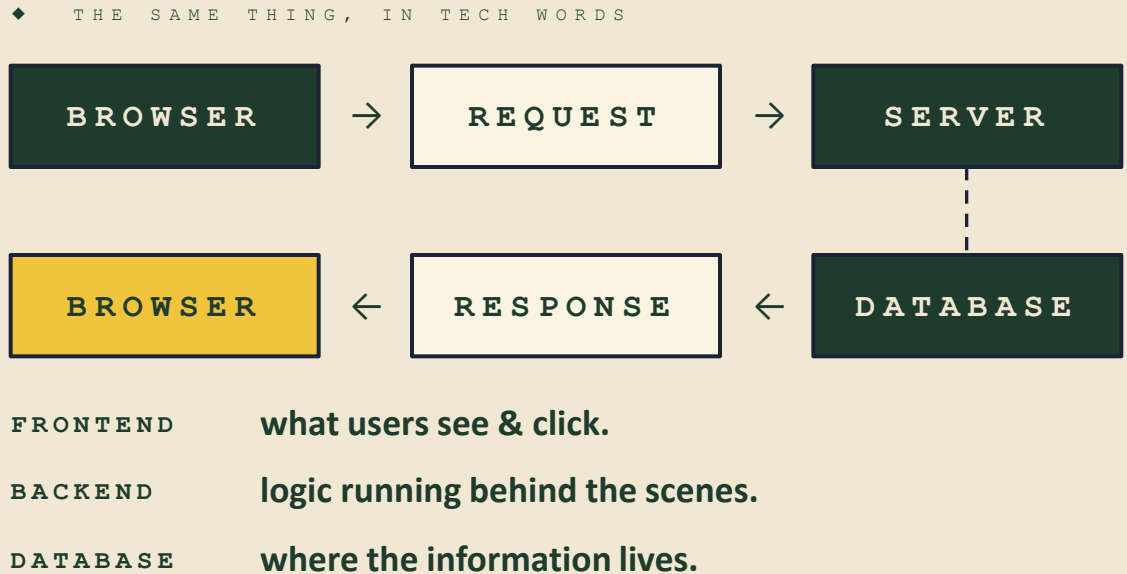
Cool. Can a browser run it?



So... what *actually* happens when you open a website?



You never walk into the kitchen yourself. That's the whole point.



ORDER TICKET #500

TABLE 7

Waiter (the API): "One user profile, please."

Kitchen (the backend): "Chef is down. Here's a 500."

...and the user just sees a sad little spinner.

Frontend is basically:

Structure + Style + Behaviour

< HTML >

STRUCTURE

"What's on the page?"



```
<button>
  Sign up
</button>
```

CSS { }

STYLE

"How does it look?"

```
button {
  background: green;
}
```



```
1 .Cat{
2   width: 100%;
3   height: 100%;
4 }
```

JAVASCRIPT () = >

BEHAVIOUR

"What happens when you click?"

```
btn.onclick = () => {
  signUp();
}
```

Why Javascript



◆ THE SAME BUTTON, THREE TIMES

Sign up

It exists. It is also ugly.

Sign up

It looks good. Still does nothing.

Sign up You're in!

It does something. Now it's a product.

HTML is the skeleton. *CSS is the skin.*

<HTML> · what is on the page

Content wrapped in tags. Each tag says what the thing is.

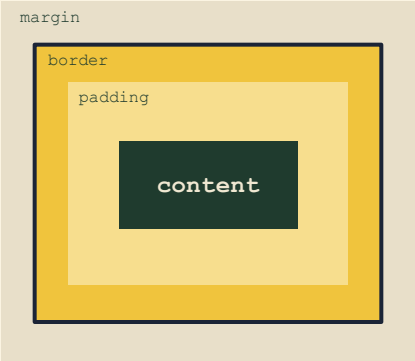
```
<header>
  <h1>Hello, judges</h1>
</header>
<main>
  <p>Built in 24 hours.</p>
  <button class="cta">Try it</button>
  
</main>
```

TAG	the label: h1, p, button, img, a
ATTRIBUTE	extra info: class, src, href, alt
SEMANTIC	header, nav, main, footer: name the parts

CSS { } · how it looks

Rules that pick an element and style it.

```
.cta {
  background: green;
  color: white;
  padding: 16px;
  border-radius: 12px;
  display: flex;
}
```



margin
border
padding
content

the box model: every element is a box

SELECTOR	who to style: .cta, button, #hero
PROPERTY	what to change: color, padding, font-size
LAYOUT	flexbox + grid handle 90% of it

JavaScript is the *brain*.

◆ THREE JOBS, ONE LANGUAGE

1

LISTEN

react to clicks, typing, scrolling (events)

```
button.onclick = ...
```

2

REMEMBER

keep data in variables and state

```
let count = 0;
```

3

TALK

ask servers for data (APIs) and show it

```
fetch("/api/profile")
```

◆ THE BUTTON FROM EARLIER, NOW ALIVE

```
const btn = document.querySelector("button");  
let count = 0;  
  
btn.onclick = () => {  
  count++;  
  btn.textContent = "Clicked " + count;  
};
```

EVENT

user clicks



CODE

count goes up



PAGE

button text updates

IDE: Has proper debugging tools
Programmer:



I have a new framework
down here

JavaScript Developers



Okay... but what *React*?

♦ HOW EVERYONE GETS THERE



♦ REACT = LEGO FOR UI



*A component is one reusable block.
Build it once, snap it in anywhere.*

```
<App />
├─ <Navbar />
├─ <Hero />
├─ <SearchBar />
├─ <Card />   × 20
├─ <Dashboard />
└─ <Footer />
```

One `<Card />` file. Twenty cards on screen.

♦ THREE WORDS, THAT'S IT

COMPONENTS
reusable pieces of UI.

PROPS
data you pass into a block.

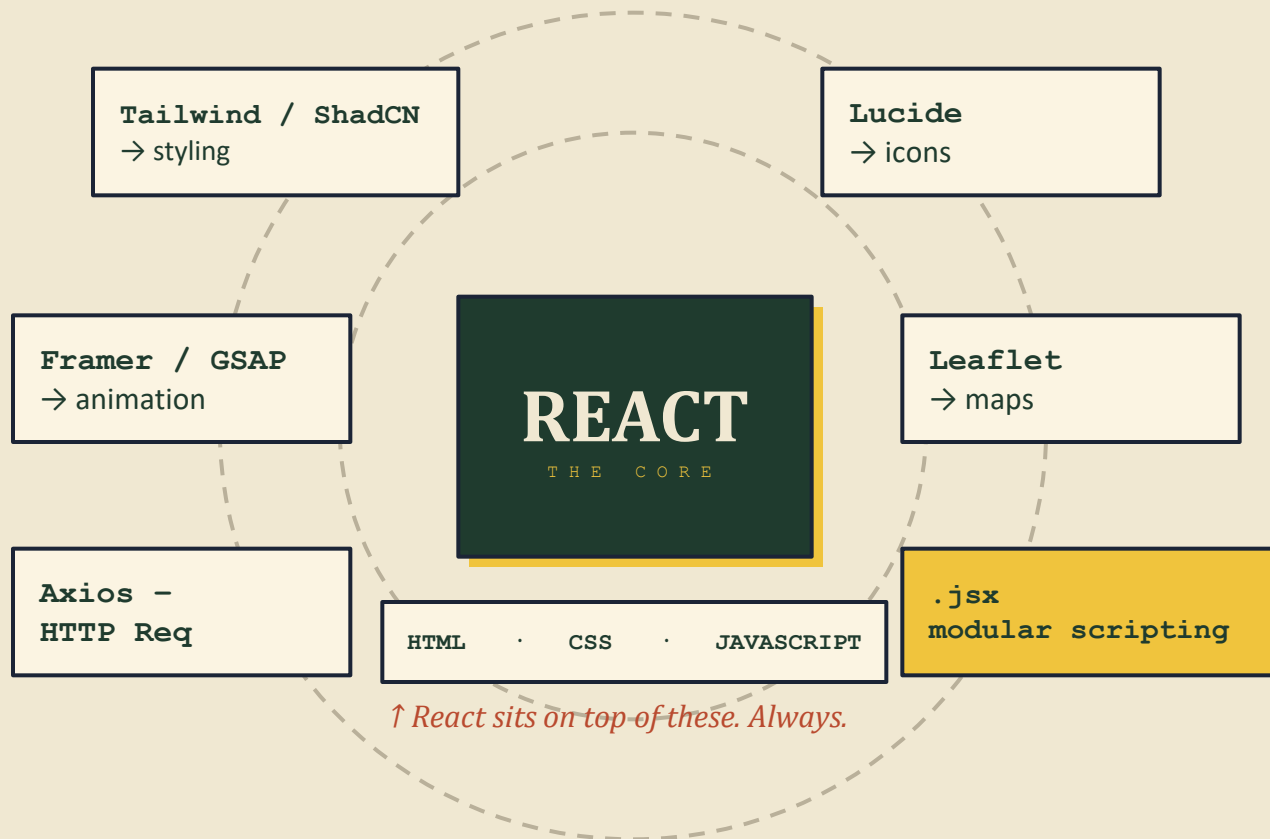
STATE
data that changes as users click.

You: "Why am I reloading the whole page after just a button update"

React: ". Stop. Just update component which changes"

Your Frontend Toolkit

Everything below is optional. Except the middle.



READ THIS TWICE

You do NOT need all of these.

◆ THE HONEST BEGINNER STACK

React

+

CSS or Tailwind

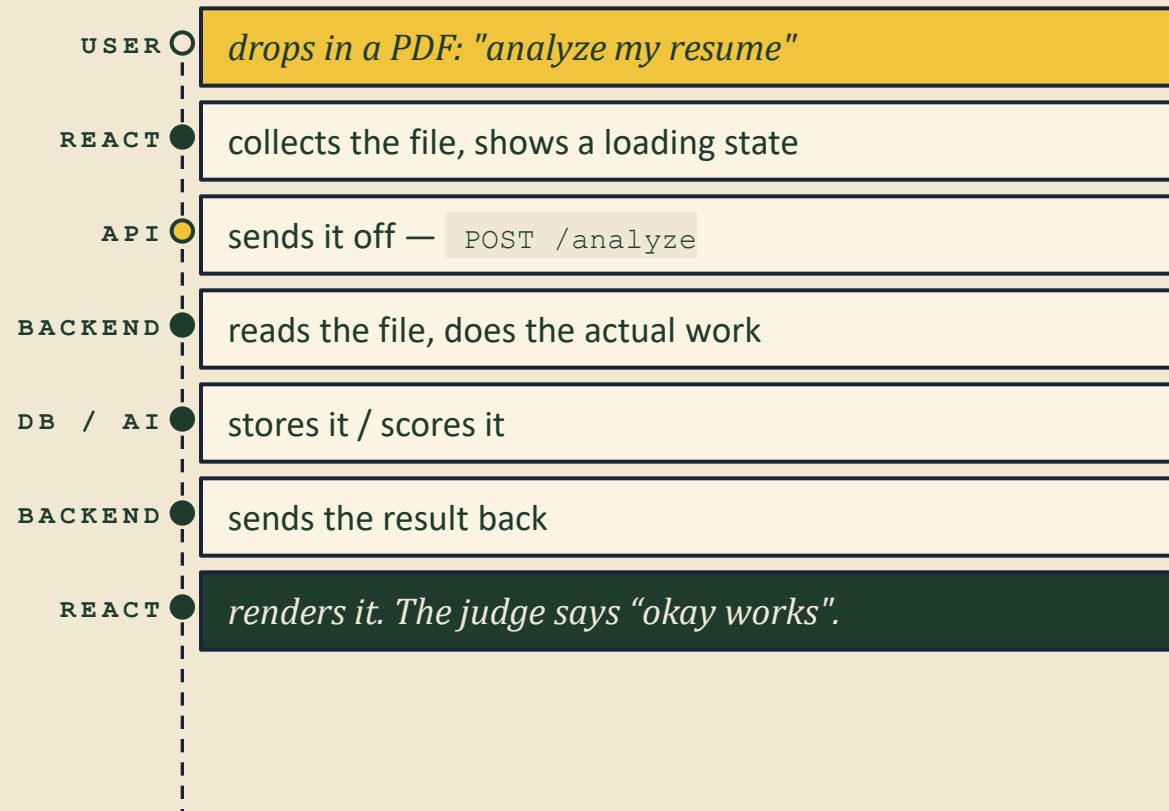
+

1-2 libraries you actually need

```
$ npm install framer-motion gsap
$ npm install ...14 more packages
...all because one button needed to bounce.
```


The frontend can't do *everything*.

◆ REAL PROJECT: "ANALYZE MY RESUME"



Backend is not magic.

It's just your code, running on a computer that isn't yours.

◆ SO WHAT IS "MERN"? FOUR NAMES.

M

MongoDB

the database — where data sleeps

E

Express

builds your API routes

R

React

the frontend — you already met it

N

Node.js

runs JavaScript on the server

*One language — JavaScript — the whole way down.
That's the entire selling point.*

REACT

— in the browser



EXPRESS + NODE

— server



MONGODB

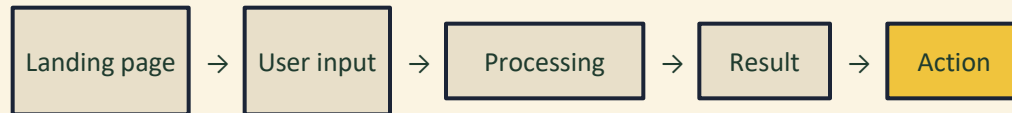
— storage

Your project only becomes real when someone else can open it.

◆ THE WHOLE WEEKEND, IN ONE LINE



BUILD ONE CORE FLOW.



One flow that works beats nine features that half-work.

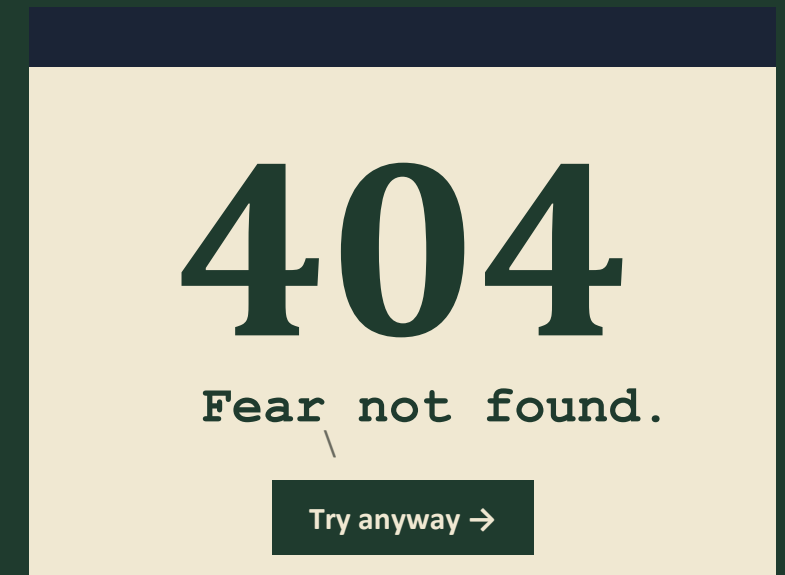
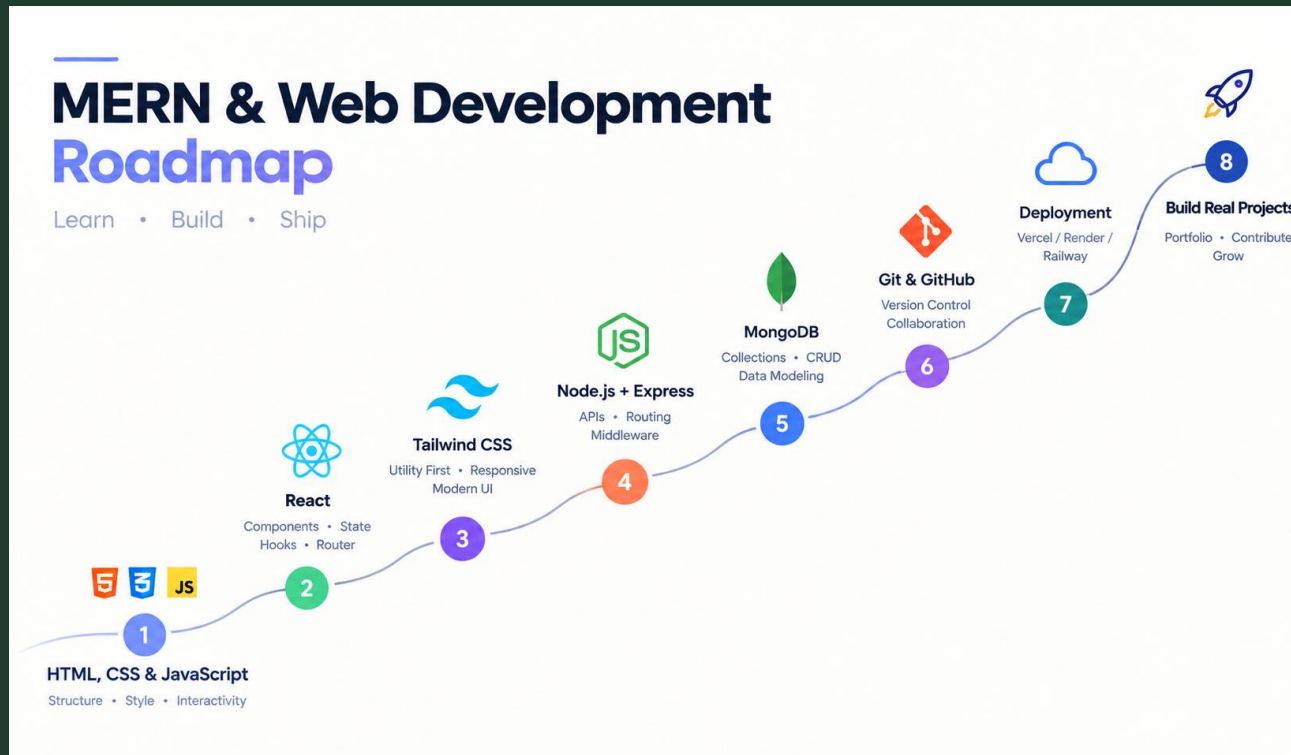
- ✗ Don't build 25 features.
- ✗ Don't spend 4 hours choosing a framework.
- ✗ Don't redesign 30 minutes before submission.

No, you cannot
work 24 hours
straight

Programmers in
Hackathon



THANK YOU. *Lets build*



You don't need to know everything.

You just need to know enough to build

IDEA → CODE → DEBUG → DEPLOY → DEMO → REPEAT